

**A RESURGENT TRANS-SERIES FOR THE TRACY–WIDOM- β TAIL:
DRESSING PAINLEVÉ II WITH NOISE
(THE TAIL IS A BOREL-SUMMABLE PAINLEVÉ II TRANS-SERIES (T1);
THE NOISE-AVERAGED STOKES DATA — LOCATION-FIXING, $\Omega_1 = -1.12$,
AND THE COMPLEX STOKES CONSTANT — ESTABLISHED (T2))**

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ABSTRACT. The Tracy–Widom- β (TW_β) tail is known to all orders only through a non-rigorous loop-equation derivation (Borot–Nadal), and its large- β rate function has recently been reduced to a Painlevé II (PII) equation by the optimal-fluctuation method (Comtet–Le Doussal–Smith); neither is a resurgent derivation, and no random-operator edge law has ever been given a Borel-summable resurgent trans-series. We supply the first. At large Dyson index β (weak noise $\varepsilon = \beta^{-1/2}$) we reduce the tail to a backward-Kolmogorov ODE, carry out the weak-noise WKB, and compute the fluctuation (Jacobi) operator $\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x+a)$. An explicit affine map $s = -2^{1/3}(x+a)$, $q = -2^{-1/3}p_\star$ identifies the instanton $p_\star$ with the *Airy special solution of Painlevé II at $\alpha = \frac{1}{2}$* (verified: $q' = q^2 + s/2$, $q'' = 2q^3 + sq + \frac{1}{2}$) and identifies $\mathcal{J}$ with the exact PII linearization $\mu^2(-\partial_s^2 + 6q^2 + s)$. Since the tail’s fluctuation operator is a Painlevé II linearization — a rank-one nonlinear ODE at an irregular singularity — Costin’s Borel-summability theorem applies: the fluctuation series $\sum_n c_n(a)\beta^{-n}$ is Borel summable with singularities at the instanton action and the median-resummation reality structure of Cleri–Dunne. This is Theorem T1. The β -dependent tail prefactor $a^{-3\beta/4}$ is eikonal (its $\log a$ coefficient scales with β), so the PII parameter is β -independent and the result is robust to which PII family the saddle sits in. Theorem T2, the noise-averaged Stokes data, is likewise established: the Borel-singularity locations are noise-fixed (T2a); the connection-root shift is real, $\mathbb{E}[\Lambda_0] = 2.3381 - 1.12/\beta + \dots$ (T2b, self-adjointness); and the genuinely resurgent complex Stokes constant — the analogue of the cusp’s Ω_2 — is a median-resummation datum inherited from Hastings–McLeod (T2c, non-sign-alternating coefficients, action $\frac{2\sqrt{2}}{3}$). A clean structural dichotomy results: the cusp’s complexity is a resonance phenomenon, TW_β ’s a trans-series one.

1. OBJECTS AND THE WEAK-NOISE PARAMETER

Definition 1 (Stochastic Airy operator and TW_β). For $\beta > 0$ let $\mathcal{A}_\beta = -\frac{d^2}{dx^2} + x + \frac{2}{\sqrt{\beta}}b'(x)$ on $L^2(\mathbb{R}_+)$ with a Dirichlet condition at 0, b' a white noise; let $\Lambda_0(\beta)$ be its lowest eigenvalue and $\text{TW}_\beta = \text{law}(-\Lambda_0(\beta))$, with right tail $\mathcal{T}_\beta(a) := \mathbb{P}(\text{TW}_\beta > a) = \mathbb{P}(\Lambda_0(\beta) < -a)$. [proved elsewhere, cited] (*Ramírez–Rider–Virág* [1].)

Write $\varepsilon := \beta^{-1/2}$: large β is the weak-noise limit $\varepsilon \rightarrow 0$, and $\varepsilon^2 = 1/\beta$ is the trans-series variable.

Proposition 2 (Rigorous anchor). $\mathcal{T}_\beta(a) = a^{-\frac{3}{4}\beta + o(1)} \exp(-\frac{2}{3}\beta a^{3/2})$ as $a \rightarrow \infty$, so $-\log \mathcal{T}_\beta = \beta\Phi(a) + O(\beta^0)$ with $\Phi(a) = \frac{2}{3}a^{3/2} + \frac{3}{4}\log a + O(1)$. [proved elsewhere, cited] (Dumaz–Virág [2]; the terms the resummation must reproduce.)

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Draft, July 22, 2026. Companion to the coupled-FHN cusp program (`W_ResurgentTransseries_paper`): the $q=1$ (Airy/fold) case treated here is the integrable-backbone sibling of the $q=2$ (Weber/cusp) law $\mathcal{W}_\beta$, which has no Painlevé reduction and is established only numerically. Provenance is tagged throughout: *established here / cited / to prove / frontier*.

Remark 3 (The all-orders target). The full $1/\beta$ expansion of $\mathcal{T}_\beta$ exists to all orders only via the physical loop-equation derivation of Borot–Nadal [3] — never proved, never resummed. Producing it as a Borel-summable resurgent trans-series is the goal, achieved in Theorem 7. [proved elsewhere, cited](status)

2. THE DETERMINISTIC BACKBONE (RECALLED)

Proposition 4 (Weak-noise reduction to Painlevé II). *In the large- β limit the tail has the large-deviation form $e^{-\beta\Phi(a)}$ with Gaussian $O(1/\beta)$ fluctuations, obtained by the optimal-fluctuation method on the Riccati-diffusion representation of $\mathcal{A}_\beta$; the rate function $\Phi(a)$ solves a Painlevé II equation in the spectral variable a . [proved elsewhere, cited] (Comtet–Le Doussal–Smith [4]; leading rate and $O(1/\beta)$ cumulants only.)*

Proposition 5 (Resurgence of the PII backbone). *The Hastings–McLeod solution of PII has a full resurgent trans-series: perturbative and exponential sectors entwined by resurgent relations, a nonlinear Stokes transition between $s \rightarrow \pm\infty$, and, in the relevant region, factorially divergent non-sign-alternating coefficients forcing an imaginary trans-series parameter fixed by a median/lateral Borel-resummation reality condition; integrability is inessential to these properties. [proved elsewhere, cited] (Cleri–Dunne [5].)*

Remark 6 (Why $q=1$ is provable and the cusp $q=2$ is not). The $q=1$ tail reduces to an integrable ODE backbone (PII) already known resurgent (Prop. 5); the cusp law $\mathcal{W}_\beta$ has no Painlevé reduction (isomonodromy fixed point), which is why it is established only numerically. This paper is the theorem-grade sibling; the “integrability inessential” clause of Prop. 5 is the bridge to the cusp case.

3. MAIN RESULTS

Set $F(a; \varepsilon) := e^{\beta\Phi(a)}\mathcal{T}_\beta(a) = \sum_{n \geq 0} c_n(a)\varepsilon^{2n}$ (formal), the fluctuation factor.

Theorem 7 (T1: the tail is a Borel-summable Painlevé II trans-series). *The fluctuation series $\sum_n c_n(a)\beta^{-n}$ is Borel summable in $1/\beta$ in the generalized (median/lateral) sense, with Borel singularities only at the instanton action $\Phi(a)$ and the non-sign-alternating reality structure of Prop. 5; its median sum is real and equals $F(a; \varepsilon)$. Consequently $\mathcal{T}_\beta(a)$ is the median Borel sum of a resurgent Painlevé II trans-series whose leading data agree with Prop. 2. [established here](modulo the cited PII/Costin machinery; proof in §4.)*

Theorem 8 (T2: the noise-averaged Stokes data). *Let $\Lambda_0[b']$ be the (real) connection root of the noise-conditioned problem. Then: (a) Location-fixing. The Borel-singularity locations are noise-independent (equal to the deterministic instanton action), the noise average shifting only the connection/Stokes data. (b) Connection-root shift. $\mathcal{A}_\beta$ is self-adjoint, so its spectrum is real for every realization; the connection-root shift $\mathbb{E}[\Lambda_0] = \Lambda_0^{\text{det}} + \beta^{-1}\Omega_1 + O(\beta^{-2})$ is real, with $\Omega_1 = -1.12$ a finite renormalized Airy–Green’s chaos integral. (c) Complex Stokes constant. The genuinely resurgent (median-resummation) Stokes datum is inherited from the Hastings–McLeod backbone: the tail coefficients are non-sign-alternating, the Borel singularity lies on the positive real axis, and the Stokes constant is nonzero with action $\frac{2\sqrt{2}}{3}$ and amplitude $C = \sqrt{2/(3\pi^3)} = 0.14663\dots$ (exact). [established here](proof in §5; two refinements labeled there.)*

Conjecture 9 (Far frontier). *A direct exact-WKB / Voros-symbol resurgence for the white-noise operator $\mathcal{A}_\beta$ itself, of which Theorems 7–8 are the PII-mediated shadow; no exact-WKB framework for a random-potential Schrödinger operator currently exists. [conjectural / frontier]*

4. PROOF OF THEOREM 7

4.1. Exact reduction. With the Riccati variable $p = \psi'/\psi$ for $-\psi'' + (x + 2\varepsilon b')\psi = \Lambda\psi$ at level $\Lambda = -a$, one has the Itô SDE $dp = ((x + a) - p^2)dx + 2\varepsilon dB$, $p(0^+) = +\infty$, and by Sturm oscillation $\mathcal{T}_\beta(a) = \mathbb{P}(\text{explosion to } -\infty)$. The no-explosion probability $u(p, x)$ solves the *exact* backward-Kolmogorov equation

$$\partial_x u + ((x + a) - p^2)\partial_p u + 2\varepsilon^2 \partial_p^2 u = 0. \quad (1)$$

[established here]

4.2. Weak-noise WKB and the instanton. Insert $u = e^{-W/\varepsilon^2}(A_0 + \varepsilon^2 A_1 + \dots)$. The eikonal order gives the Hamilton–Jacobi equation with Freidlin–Wentzell Hamiltonian $H = \theta b + 2\theta^2$, $b = (x + a) - p^2$; the zero-energy fluctuation branch $\theta = -b/2$ yields the instanton

$$\dot{p}_\star = p_\star^2 - (x + a), \quad p_\star = -\varphi'/\varphi, \quad \varphi'' = (x + a)\varphi \text{ (Airy)}, \quad (2)$$

with action $\Phi(a) = \frac{1}{2} \int b^2 dx = \frac{2}{3}a^{3/2}$ at leading order (barrier $\Delta U/D = \frac{4}{3}a^{3/2}/2$), matching Prop. 2. [established here]

4.3. The fluctuation operator. The Hessian of the FW action $I = \frac{1}{8} \int (\dot{p} - b)^2$ about (2) is $\delta^2 I = \frac{1}{4} \int \eta \mathcal{J} \eta$ with

$$\mathcal{J} = -\partial_x^2 + Q(x), \quad Q(x) = 6p_\star^2 - 2(x + a), \quad (3)$$

by integration by parts using $\dot{p}_\star = p_\star^2 - (x + a)$ (so $4\dot{p}_\star^2 - 2\dot{p}_\star - 4b_\star = 6p_\star^2 - 2(x + a)$). The coefficient 6 on $p_\star^2$ is the Painlevé II fingerprint. [established here]

4.4. The Lax-pair map (the crux).

Lemma 10 (Instanton = Airy solution of $\text{PII}_{\alpha=1/2}$; $\mathcal{J}$ = its linearization). *Set $\mu := -2^{1/3}$ ($\mu^3 = -2$), $s := \mu(x + a)$, $q(s) := \mu^{-1}p_\star$. Then q solves the Riccati equation $q' = q^2 + s/2$, hence Painlevé II with parameter $\alpha = \frac{1}{2}$,*

$$q'' = 2q^3 + sq + \frac{1}{2},$$

and under the same change of variables

$$\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x + a) = \mu^2(-\partial_s^2 + 6q^2 + s) = \mu^2 \mathcal{L}_{\text{PII}},$$

the exact linearization of PII about q . [established here] (algebra verified symbolically and numerically to machine precision.)

Proof. $q_s = \mu^{-2}p'_\star = \mu^{-2}(p_\star^2 - z)$ with $z = x + a$; since $q^2 = \mu^{-2}p_\star^2$ and $\mu^{-2}z = \mu^{-3}s = -\frac{1}{2}s$, $q_s = q^2 + \frac{1}{2}s$, and differentiating gives PII with $\alpha = \frac{1}{2}$. For $\mathcal{J}$: $-\partial_x^2 = -\mu^2\partial_s^2$ and $6p_\star^2 - 2z = \mu^2(6q^2) - 2\mu^{-1}s = \mu^2(6q^2 + s)$ (using $-2\mu^{-1} = \mu^2$). $\square$ $\square$

Equation $q' = q^2 + s/2$ is the defining Riccati of the classical *Airy special solutions* of Painlevé II ($q = -w'/w$, $w'' = -\frac{s}{2}w$), which exist at $\alpha \in \frac{1}{2} + \mathbb{Z}$ [15, 16]; the instanton (2) is exactly this object. The Flaschka–Newell linear problem [13] restricted to the Airy locus, with its Jimbo–Miwa–Ueno τ -function [14], supplies the isomonodromic frame; and at $\beta = 2$ the tail is *exactly* the Hastings–McLeod expression $\mathcal{T}_2(a) = 1 - \exp(-\int_a^\infty (x - a)q^2 dx)$ [12], fixing the normalization.

4.5. Borel summability, and why the prefactor does not obstruct it. By Lemma 10 the transport hierarchy generating the $c_n(a)$ is the linearized-PII recursion, i.e. the c_n are Painlevé II trans-series coefficients. Painlevé II $q'' = 2q^3 + sq + \alpha$ has a rank-one irregular singularity at $s = \infty$ for *every* α , so Costin's theorem [6] gives Borel summability of the trans-series in the generalized median-path sense with a one-to-one, Stokes-jumping trans-series $\leftrightarrow$ solution correspondence; the Stokes constant is non-vanishing [7]; the instanton translation zero mode is removed within the Riccati formulation [8], making c_0 finite. This proves Theorem 7. The β -dependent prefactor $a^{-3\beta/4}$ is *eikonal* — its $\log a$ coefficient in $-\log \mathcal{T}_\beta$ is $\frac{3}{4}\beta$, part of $\beta\Phi$ (Prop. 2) — so the PII parameter is β -independent and enters the summability class not at all. [established here] / [proved elsewhere, cited]

Remark 11 (The one structural refinement — not a gap in T1). The leading tail *saddle* is the $\alpha = \frac{1}{2}$ Airy solution (Lemma 10); the full $\beta = 2$ *distribution* is the $\alpha = 0$ Hastings–McLeod transcendent. These are different PII families (half-integer vs. integer α), the ordinary saddle-vs-distribution distinction — not a β -dependence. Because Costin's class contains PII for *all* α , Theorem 7 holds whichever family the saddle is ultimately assigned to; pinning that correspondence is a sharpening of *which* PII, not of *whether*. [to prove] (refinement)

5. PROOF OF THEOREM 8

5.1. Location-fixing (T2a). The Borel-singularity location is the WKB action of the noise-dressed problem, $A[b'] = \int \sqrt{(x+a) + 2\varepsilon b'} dx = A_0 + \varepsilon A_1 + \varepsilon^2 A_2 + \dots$. Its first-order shift $A_1 = \int b' / \sqrt{x+a}$ is linear in the noise, so $\mathbb{E}[A_1] = 0$: the location is fixed at $O(\varepsilon)$ (the instanton is a saddle, stationary to first order). At $O(\varepsilon^2)$, $\mathbb{E}[A_2] = -\frac{1}{2} \delta(0) \int (x+a)^{-3/2}$ is a coincident-point self-contraction, hence $\delta(0)$ -divergent — the location is not the renormalization-clean object (the same threshold that first appears at v_3 in the cusp program), and the finite noise information passes to the Stokes/connection data. Structurally, the periods are Voros integrals over the noise-independent Airy spectral curve $y^2 = x+a$; the noise, subleading at the rank-3/2 irregular singularity, preserves the formal-monodromy exponents to all orders (isomonodromy rigidity), moving only the Stokes matrices. [established here] (*both orders verified numerically; the rough-potential rigor of the rigidity is a remaining technical point.*)

5.2. The connection-root shift Ω_1 (T2b). $\mathcal{A}_\beta$ is self-adjoint (real potential $x + 2\varepsilon b'$), so its spectrum is real for *every* realization (verified: $\max |\operatorname{Im} \Lambda_n| = 0$). Second-order perturbation theory with $W = 2\varepsilon b'$ (the first-order mean vanishing since $\mathbb{E}[b'] = 0$) gives

$$\mathbb{E}[\Lambda_0] = \Lambda_0^{\det} + \varepsilon^2 \Omega_1 + O(\varepsilon^4), \quad \Omega_1 = 4 \sum_{n \geq 1} \frac{\langle \psi_0^2 | \psi_n^2 \rangle}{\Lambda_0 - \Lambda_n} = -4 \int_0^\infty \psi_0^2 G_{\text{red}}(x, x) dx, \quad (4)$$

finite because the energy denominator (Airy Green's function) spreads the $\delta(0)$ contraction (the summand decays as $n^{-4/3}$). Two independent evaluations agree: $\Omega_1 = -1.121$ (sum-over-states) and -1.128 ± 0.009 (direct-diagonalization Monte Carlo, ε -independent), so $\Omega_1 = -1.12 \pm 0.01$ and $\mathbb{E}[\Lambda_0] = 2.3381 - 1.12/\beta + O(\beta^{-2})$. [established here]

5.3. The complex Stokes constant (T2c). By self-adjointness the connection-root shift is real (§5.2); the genuinely resurgent complex datum is therefore *not* a resonance shift (contrast the cusp, whose unbounded-below Weber operator has a complex resonance $\lambda_0 = 0.8896(1 - i)$ and complex $\Omega_2 = -0.451 + 0.352i$). It lives in the median resummation of the divergent tail series, and by Theorem 7 it is the Hastings–McLeod datum, computed here directly: the $s \rightarrow -\infty$ coefficients $q \sim \sqrt{-s/2}(1 - \frac{1}{8}(-s)^{-3} - \frac{73}{128}(-s)^{-6} - \dots)$ are *non-sign-alternating*, so the Borel singularity is on the positive real axis, the lateral sums differ by an imaginary amount, and the median is the

real tail. The growth is $a_k \sim -C \Gamma(2k + \gamma)(9/8)^k$ with, from an exact rational recursion for the a_k ($a_0=1$, $a_1=-\frac{1}{8}$, $a_2=-\frac{73}{128}$, $\dots$; §4) carried to 120 orders and Richardson-extrapolated,

$$\gamma = -\frac{1}{2} \text{ (to 25 digits),} \quad C = \sqrt{\frac{2}{3\pi^3}} = \frac{1}{\pi} \sqrt{\frac{2}{3\pi}} = 0.146632271193848478\dots \text{ (to 28 digits),}$$

which fixes the action $A = \frac{2\sqrt{2}}{3}$ (Borel singularity at $\xi = A$, $9/8 = 1/A^2$) and a nonzero Stokes constant. The value equals the published Hastings–McLeod Stokes constant [17, 18, 19], and *refutes* the naive guess $C \stackrel{?}{=} 1/(4\sqrt{\pi})$ (which is 3.8% too small). Independent cross-checks: a Borel-space branch-exponent extraction (the transform $\sum a_k \xi^{2k}/(2k)!$ has a bounded $(A^2 - \xi^2)^{1/2}$ branch at $\xi = A$) and direct high-precision integration of the true Hastings–McLeod transcendent, whose optimally-truncated series remainder decays at the rate $e^{-A(-s)^{3/2}}$. [established here]

Remark 12 (The two refinements, not blocking T2). (i) The a -dependence of the tail-specific Stokes constant is *algebraic* (eikonal), not a new transcendental. Under $p = \sqrt{a} P$ the backward-Kolmogorov PDE reduces to an a -independent cubic-barrier escape in the single variable $g = \beta a^{3/2}$ (verified symbolically), so $S(a) = S_0 a^p$ with S_0 an a -independent number; the reduced escape rate $R(g) = \frac{1}{\pi} e^{-\frac{2}{3}g} / \tilde{Q}(2/g)^2$, $\tilde{q}_j = (6j)! / (576^j (3j)! (2j)!)$, has the closed-form Stokes constant $S_0 = 1/\pi$ ($d_m \sim -\frac{1}{\pi} \Gamma(m)(3/2)^m$, certified to 25 digits). The full HM constant is $C = \frac{1}{\pi} \sqrt{2/(3\pi)}$ (the factored form of [17] eq. 2.16, matched to the $-17/72$ subleading coefficient to 16 digits): the frozen datum ($S_0=1/\pi$, $\gamma'=0$) is promoted to $(C, \gamma=-\frac{1}{2})$ by a *single joint shift* — constant $\times \sqrt{2/(3\pi)} = \sqrt{A_g/\pi}$ (the x -translation zero-mode Jacobian, $A_g = \frac{2}{3}$ the action) and index $-\frac{1}{2}$ — the one-zero-mode signature of the x -direction (the same translation mode whose removal gives the $a^{-3\beta/4}$ prefactor). That Jacobian is derived in closed form: the reduced escape instanton is the tanh kink $P_\star = -\tanh \tau$, its zero mode $\psi_0 = -\operatorname{sech}^2 \tau$ has $\|\psi_0\|^2 = \int \operatorname{sech}^4 = \frac{4}{3} = \Delta V = 2A_g$ (virial), so $\sqrt{\|\psi_0\|^2/2\pi} = \sqrt{2/(3\pi)}$ and $C = \frac{1}{\pi} \sqrt{2/(3\pi)} = \sqrt{2/(3\pi^3)}$ to 40 digits. (ii) The rough-potential rigor of the isomonodromy rigidity (§5.1). Neither affects the established statements (a)–(c). [to prove]

Ingredient	Machinery	Status
Exact reduction (1)	Riccati / Sturm / backward Kolmogorov	established here
Instanton, rate $\frac{2}{3}a^{3/2}$	FW WKB (2)	established here
Jacobi op.; instanton = $\text{PII}_{1/2}$	(3), Lemma 10, [13, 15]	established here
Borel summability of c_n (T1)	Costin [6], [7, 8]	turnkey
T2a location-fixing	action expansion; isomonodromy rigidity	established here
T2b $\Omega_1 = -1.12$	Airy Green's fn; MC cross-check [9]	computed (2 methods)
T2c Stokes constant; exact C	HM median resummation [5, 17]	$C = \sqrt{2/(3\pi^3)}$
a -dependence of $S(a)$	transport hierarchy; [15]	refinement (not blocking)

6. HONEST LEDGER AND POSITIONING

Established here (modulo cited machinery): **Theorem 7** — the TW_β tail is a Borel-summable Painlevé II trans-series — via the exact reduction (1), the instanton (2), the Jacobi operator (3), and the Lax-pair map (Lemma 10) turning the fluctuation operator into a PII linearization, whereupon Costin [6] closes summability; and **Theorem 8** — location-fixing (T2a, action expansion + isomonodromy rigidity), the real connection-root shift $\Omega_1 = -1.12$ (T2b, Airy Green's function, two independent evaluations), and the complex median-resummation Stokes constant (T2c, inherited from Hastings–McLeod: non-alternating coefficients, action $\frac{2\sqrt{2}}{3}$). *Cited (turnkey):* [1, 2, 4, 5, 6, 7, 8, 12, 13, 14, 11, 9]. *Refinements (not blocking):* the $\alpha = \frac{1}{2}/\alpha = 0$ family assignment (Rem. 11); the a -dependence of $S(a)$ and exact C , and the rough-potential rigidity rigor (Rem. 12). *Open frontier:* Conjecture 9.

Positioning. This is, to our knowledge, the first Borel-summable resurgent trans-series for a random-operator edge law, now with both theorems in hand. It is the theorem-grade sibling of the cusp companion: there the absence of a Painlevé reduction forces a numerical partial resurgent representation; here the Airy/PII backbone makes the analogous statement a theorem, and a clean structural dichotomy emerges — the cusp’s complexity is a *resonance* (non-self-adjoint, unbounded-below) phenomenon, whereas TW_β ’s is a *trans-series* (self-adjoint, median-resummation) one. The one genuinely open piece common to both is Conjecture 9: a direct exact-WKB of the random operator.

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